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NAAC GRADE A+
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 6

Title: Sorting and Filtering Data in Power BI

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Experiment No.: 6

Sorting and Filtering Data in Power BI

GitHub-Repo-Link: prakruthidevanga/Business-Analytics: Business Analytics

1. Problem Statement

A sales manager wants to identify the company's highest-value transactions and underperforming products. As a data analyst, I used sorting and filtering techniques in Power BI to extract the relevant business information from the sales dataset. The dataset covers October 2025 to September 2026 and contains 300 transactions across 12 products, 4 regions and 8 salespeople.

2. Methodology / Steps Followed

Step 1: Data Collection

- Obtained the sales dataset in Excel format (Sales_Analysis_Dataset.xlsx).
- Used the Sales_Data sheet, which contains Transaction ID, Date, Region, Salesperson, Customer, Product, Category, Quantity, Unit Price, Discount %, Unit Cost, Revenue, Profit and Returned.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported the Sales_Data sheet using **Get Data → Excel workbook**.
- Loaded the table using **Transform Data** and **Close & Apply**.

Step 3: Data Cleaning and Transformation

- Checked for duplicate records and missing values. None were found, and all 300 Transaction IDs are unique.
- Set the correct data types: Date as Date, Quantity as Whole Number, Unit Price, Unit Cost, Revenue and Profit as Decimal number, and Discount % as Percentage.

Step 4: Creating DAX Measures

- Created measures for Total Revenue, Total Profit, Profit Margin % and Units Sold.
- Created a High Value Count measure: the number of transactions with Revenue ≥ $32,400.
- Created an Underperforming Flag measure to label each product as "Underperforming" or "OK". [Edit this line to state your own rule, for example: a product is flagged when its profit margin is below 15% or its revenue is far below the product average.]

Step 5: Sorting and Filtering

- **Sorting:** Sorted the transactions table by Revenue in descending order to rank the highest-value transactions. Sorted the product table by Total Revenue in ascending order to bring the weakest products to the top.
- **Top N filter:** Applied a Top 15 filter on Revenue to list the 15 largest transactions.
- **Visual-level filter:** Filtered the Underperforming Flag to show underperforming products.
- **Slicers:** Added slicers for Region, Category and Date.

Step 6: Dashboard Development

- Designed a one-page dashboard titled “High-Value Transactions and Underperforming Products Analysis”.
- Added KPI cards for Total Profit, Total Revenue, Profit Margin % and High Value Count.
- Added a Top 15 transactions table and bar charts for Total Revenue by Product and Total Profit by Product.
- Added a product table with red conditional formatting for underperforming products.
- Added Region, Category and Date slicers so the data can be explored interactively.

3. Output

The final Power BI dashboard provides:

- Key figures: Total Revenue of $4.29M, Total Profit of $1.71M and a profit margin of 39.8%.
- A ranked list of the 15 highest-value transactions.
- A count of high-value transactions (32).
- Product-wise revenue and profit comparison.
- Highlighted underperforming products.
- Interactive filters by region, category and date.

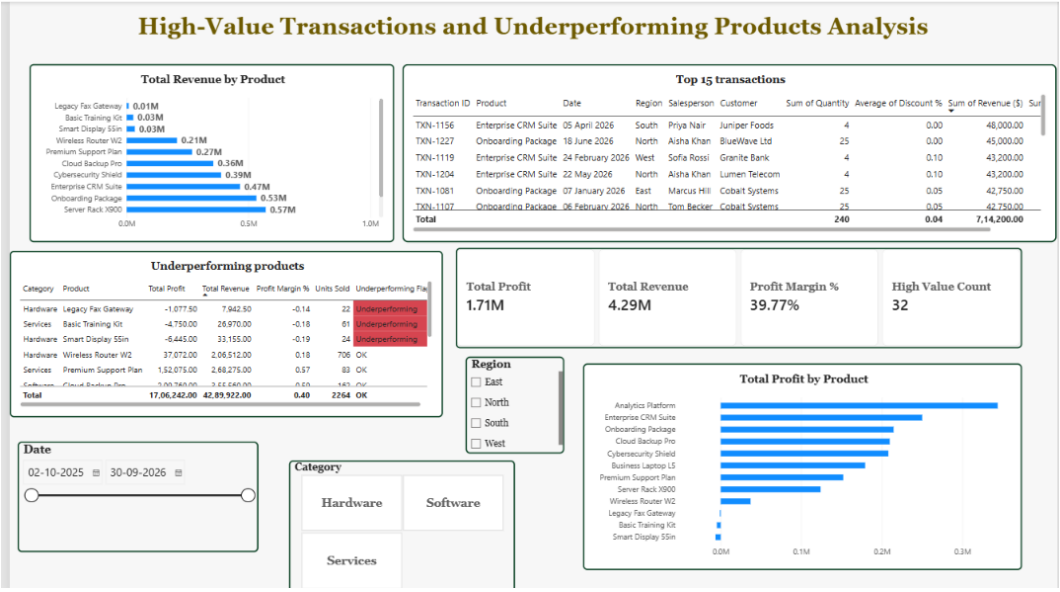


Table: Underperforming products

Product	Category	Units Sold	Revenue	Profit	Margin
Legacy Fax Gateway	Hardware	22	$7,942.50	−$1,077.50	−13.6%
Basic Training Kit	Services	61	$26,970	−$4,750	−17.6%
Smart Display 55in	Hardware	24	$33,155	−$6,445	−19.4%

4. Insights & Findings

- The Top 15 transactions total $714,200 in revenue and $305,300 in profit, about 16.6% of total revenue.
- There are 32 high-value transactions (Revenue ≥ $32,400). They are about 11% of all transactions but generate about $1.19M, or 28% of total revenue.
- Enterprise CRM Suite and Onboarding Package dominate the highest-value deals.
- Analytics Platform and Enterprise CRM Suite earn the highest profit among products.
- Legacy Fax Gateway, Basic Training Kit and Smart Display 55in have both low revenue and negative profit.
- Wireless Router W2 is profitable (18% margin) but has comparatively low revenue ($0.21M).

5. Conclusion

This experiment used sorting and filtering in Power BI to identify high-value transactions and underperforming products. Sorting by revenue ranked the largest deals, and Top N and advanced filters isolated the 15 largest and the 32 high-value transactions. A flag-based filter isolated three loss-making products. The dashboard gives the sales manager an interactive view of the region, category and date. Management can focus on the high-value product lines and review the pricing, discounts or continuation of the loss-making products.

6. Learning Outcomes

- How to import an Excel dataset into Power BI and set data types.
- How to sort data in ascending and descending order.
- How to apply Top N, advanced and visual-level filters.
- How to write DAX measures such as Total Revenue, Profit Margin % and a conditional flag.
- How to use slicers and conditional formatting to build an interactive dashboard.
- How to turn raw sales data into business insights.